

Exam Date & Time: 06-Mar-2025 (08:30 AM - 10:00 AM)



MANIPAL ACADEMY OF HIGHER EDUCATION

ENGINEERING MATHEMATICS - IV [MAT 2226]

Marks: 30

Duration: 90 mins.

MCQ

Answer all the questions.

Section Duration: 20 mins

- 1) Let X be the random variable with expectation 10 and variance 9. Then the lower bound to $P(4 < X < 16)$ is...

1)	0.75		2)	0.69		3)	0.96		4)	0	
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(0.5)

Correct option is: 1

- 2) If the random variable X has pdf $f(x) = \begin{cases} 2x; & 0 \leq x \leq 1 \\ 0; & \text{Otherwise} \end{cases}$ then Median of X is...

1)	$\frac{1}{\sqrt{2}}$		2)	$\frac{2}{3}$		3)	$\frac{1}{3}$		4)	$\frac{1}{2}$	
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(0.5)

Correct option is: 1

- 3) Let X and Y be two independent random variables such that $0 \leq X, Y \leq 2$. If $P[X = 0] = P[Y = 0] = 1/2$, $P[X = 1] = P[Y = 1] = 1/3$, and $P[X = 2] = P[Y = 2] = 1/6$. Then $P[X=2, Y=1] + P[X=0, Y=1] = \dots$

1)	$\frac{1}{6}$		2)	$\frac{1}{18}$		3)	$\frac{2}{9}$		4)	$\frac{11}{18}$	
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(0.5)

Correct option is: 3

- 4) A biased coin lands on heads with probability p and on tails with probability $1 - p$. You flip the coin twice. Given that at least one flip was heads, what is the probability that the first flip was heads?

1)	p		2)	$\frac{p}{1-p^2}$		3)	$\frac{p}{p+(1-p)^2}$		4)	$\frac{p}{p+(1-p)p}$	
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(0.5)

Correct option is: 4

- 5) If X is a Poisson variate such that $E(X^2) = 6$ then $E(X) = \dots$

1)	-3		2)	$\sqrt{6}$		3)	3		4)	2	
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(0.5)

Correct option is: 4

- 6) Suppose $A_1, A_2, \dots, A_n$ are independent events each with probability p , then what is the probability that at least one of the events will occur?

1)	$1 + (1 + p)^n$		2)	$1 - (1 + p)^n$		3)	$1 + (1 - p)^n$		4)	$1 - (1 - p)^n$	
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(0.5)

Correct option is: 4

- 7) A bag contains 3 red, 2 white and 4 green balls. Two balls are drawn from the bag one after the other with replacement. The probability that the first ball drawn is green and the second ball drawn is white is...

1)	$\frac{1}{9}$		2)	$\frac{2}{9}$		3)	$\frac{2}{81}$		4)	$\frac{8}{81}$	
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(0.5)

Correct option is: 4

- 8) If X is a Binomially distributed random variable with $E(X) = 2$ and $V(X) = \frac{4}{3}$ then number of trials $n = \dots$

1)	3		2)	9		3)	8		4)	6	
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(0.5)

Correct option is: 4

- 9) Let X and Y be two independent random variables such that $P(X) = 0.40$ and $P(X \cup \bar{Y}) = 0.7$. Then the value of $P(\bar{Y})$ is...

1)	0.7		2)	0.5		3)	0.4		4)	0.3	
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(0.5)

Correct option is: 2

- 10) Which of the following is NOT a correct statement?

(0.5)

1)	$0 \leq P(A) \leq 1$, for any event A in a sample space S		2)	$P(\emptyset) = 0$		3)	If A and B are mutually exclusive events in a sample space S , then $P(A \cap B) = P(A) \cdot P(B)$		4)	If S is a sample space, then $P(S) = 1$	
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Correct option is: 3

DESCRIPTIVE

Answer all the questions.

- 11) A company has three production units: Unit X, Unit Y, and Unit Z. Unit X produces 40% of the total goods, Unit Y produces 35%, and Unit Z produces the rest. A defect product occurs in 2% of the goods produced by Unit X, 3% of the goods produced by Unit Y, and 5% of the goods produced by Unit Z. A product is randomly selected, and it turns out to be defective. (4)
- (a) What is the probability that it was produced by Unit Y?
- (b) Are the events "The product is produced by Unit Y" and "The product is defective" independent? Justify mathematically.
- 12) Suppose that machine 2 produces twice as many items as produced (per day) by machine 1. About 4% of the items from machine 1 tends to be defective while machine 2 produces 2% defective items. Suppose that the daily output of the two machines is combined and 10 items are taken from the combined output. What is the probability that 2 defective items are found? (3)
- 13) Two dimensional random variable (X, Y) with joint pdf
- $$f(x, y) = \begin{cases} 6e^{-2x-3y}; & x, y > 0 \\ 0 & ; \text{elsewhere} \end{cases}$$
- Then find (3)
- (a) Marginal probability density function of X
- (b) Marginal probability density function of Y
- (c) Conditional probability density function of X given Y.
- 14) Four balls are randomly selected from an urn containing 4 white and 4 red balls. If white ball is selected then the person wins \$5 and if red ball is selected then the person loses \$5. Let X be total winning from the experiment. Determine the probability distribution of X and calculate Variance of X. (3)
- 15) If X_1, X_2, X_3 are uncorrelated random variables with $\sigma = 5, 10$ and 24 respectively then find the correlation coefficient between U and V, where $U = X_1 + X_2$ and $V = X_1 - X_3$. (3)
- 16) Find the mean and variance of a continuous random variable X having the pdf
- $$f(x) = \begin{cases} xe^{\frac{-x^2}{2}}, & x > 0 \\ 0, & \text{Otherwise} \end{cases}$$
- (3)
- 17) A six-sided die is designed in such a way that the probability of occurrence of each face is proportional to the number on that face and is rolled once. Then find the probability that the outcome is (2)
- (a) an odd number
- (b) an odd number or a composite number

- 18) A and B throw alternately with a pair of dice, and the sum of the two outcomes is noted. The player who obtains 7 first wins. If A begins, what is the probability that A wins? (2)
- 19) If the two events A and B are independent, then verify whether A and $\bar{B}$ are independent? (2)

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